A certified lower bound on the topological entropy of the classical Hénon map: covering relations composed to an exact integer spectral argument

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Abstract

For the Hénon map $F(x, y) = (1 - ax^2 + by,\ x)$ at the classical parameters $a = 1.4$, $b = 0.3$ we prove $h_{\mathrm{top}}(F) \geq \frac{1}{88} \ln 338{,}518{,}196{,}978 > 0.301680$. The proof is a finite certificate: 340 pairwise disjoint parallelograms (h-sets) in the plane; 4,140 covering relations $N_i \Rightarrow N_j$ for the iterates F^k, $k = 1, \ldots, 6$, each decided as a finite set of strict inequalities in outward-rounded interval arithmetic with adaptive bisection; the relations chained to the uniform iterate F^{11} as a binary relation on the h-sets; and an exact integer bound on the spectral radius of that relation (the minimum row sum of its 8th power on a 223-vertex core is 338,518,196,978). The one external theorem consumed is the covering-relation theorem of Zgliczyński and Gidea. The instrument is calibrated where the answer is known: at $a = 6$ the same code certifies the full two-shift and returns $\ln 2$. The number is not claimed to improve any published bound. What is offered is the decision grammar—every step either certifies or refuses, and no floating-point quantity decides anything—and a public record that a 65-second battery re-proves in full, the certificate included. Two defects of earlier versions of the verifier, each found because it produced a bound above the value the literature reports for this map, are described together with what now guards them; the interim bound 0.356403 that one of them produced is named as tainted and withdrawn. The certified cycle counts of the same laboratory put the growth rate of periodic points at 0.4658, and the distance to it is stated as the open problem, with the cheapest next step identified in the certificate itself.

1 Introduction

Hénon introduced the map $F_{a,b}(x, y) = (1 - ax^2 + by,\ x)$ in 1976 as the simplest polynomial diffeomorphism of the plane with a numerically observed strange attractor, at the parameters $a = 1.4$, $b = 0.3$ [1]. Fifty years later the attractor at those parameters is still the standard test object of two-dimensional chaos, and the simplest quantitative statement one can make about it, the topological entropy $h_{\mathrm{top}}(F)$ of Adler, Konheim and McAndrew [2]—the exponential growth rate of the number of orbit segments distinguishable at a fixed resolution—is not known in closed

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

form. The report that accompanies the record behind this paper quotes the value the literature reports as approximately 0.4651 without naming its source; this draft repeats neither the value's provenance nor any published rigorous bound, because none is pinned in the record from which every number here is interpolated. Rigorous lower bounds for this map by covering relations exist in the literature and the author's understanding is that they exceed the bound proved here; the comparison is the first item for reviewers (Section 8).

This paper proves one inequality, $h_{\mathrm{top}}(F) \geq \frac{1}{88} \ln 338{,}518{,}196{,}978$, and its contribution is not the inequality. The method is the covering-relations method of Zgliczyński and Gidea [3], the standard tool for computer-assisted entropy bounds in low dimension. What is new here is the decision grammar and the public re-runnable record:

- every covering relation is decided as a finite set of strict interval inequalities, by adaptive bisection with outward rounding, with refusal (budget exhausted, condition not resolved) as the only alternative to certification, so that the verifier can refuse a true relation but cannot certify a false one;
- relations of different durations are chained to one uniform iterate as a *binary* relation on the h-sets, so that no path multiplicity enters the count of itineraries;
- the spectral radius is bounded from below in exact integer arithmetic, by a row-sum argument, never by a floating-point eigenvalue;
- the certificate is finite data (parallelogram matrices, an edge list) in a public file, and a battery re-derives every relation, the disjointness of every pair and the bound in 65 seconds at every build of the repository and of this document.

The instrument was calibrated before any new number counted: at $a = 6$, deep in the horseshoe regime of Devaney and Nitecki [4], it certifies the full two-shift and returns $\ln 2$ (Proposition 1). Two earlier versions of the verifier were wrong, and both were found the same way: they returned a bound above the value the literature reports for this map, which is impossible for a lower bound that is sound. Section 6 describes both defects as engineering of the verifier, names the interim number one of them produced, and states what guards each today, including one guard that, as it runs, does not exercise what it was written to pin.

A word on vocabulary. *Certified* is used in one sense only: a statement is certified when a finite set of inequalities whose truth implies it has been evaluated in outward-rounded interval arithmetic, or in exact integer arithmetic, and found to hold. Elsewhere we say *decided* or *proved*.

2 Statement of results

Throughout, $F = F_{a,b}$ with a and b the IEEE-754 doubles nearest 1.4 and 0.3,

$$a = 1.3999999999999999112\ldots, \qquad b = 0.29999999999999998890\ldots,$$

which are the objects the record is stated for (Remark 2 extends this). F is a diffeomorphism of the plane since $b \neq 0$. For a compact F-invariant set Λ we write $h_{\mathrm{top}}(F|_\Lambda)$ for the topological entropy of the restriction; $h_{\mathrm{top}}(F)$ denotes the supremum over compact invariant sets, which is the entropy of the restriction to the nonwandering set when that set is compact, and every lower bound below is a bound on $h_{\mathrm{top}}(F|_\Lambda)$ for an explicit compact invariant Λ, so any convention monotone in compact invariant subsets receives it.

Theorem 1.

$$h_{\mathrm{top}}(F) \;\geq\; \tfrac{1}{88}\ln 338{,}518{,}196{,}978 \;=\; 0.301680041\ldots \;>\; 0.301680.$$

The bound is an explicit real number: the 8th root of the integer 338,518,196,978 is a lower bound for the spectral radius of a 223×223 zero–one matrix read off the certificate, and that matrix encodes the uniform iterate F^{11}; the decimal is the floating-point evaluation of $\ln(338518196978)/88$, truncated.

Remark (the parameters)**.** The record is stated for the doubles because the verifier takes its parameters as point intervals. The same verifier accepts parameter intervals, and while this draft was prepared it was re-run with a and b replaced by the one-ulp intervals

$$a \in [1.3999999999999996891\ldots,\; 1.4000000000000001332\ldots],$$
$$b \in [0.29999999999999993339\ldots,\; 0.30000000000000004441\ldots],$$

which contain the rationals $7/5$ and $3/10$ (decided exactly, by cross-multiplication of the doubles' dyadic expansions): all 4,140 relations re-certified and the bound is unchanged (0.301680). Since nothing else in the proof depends on the parameters, Theorem 1 holds for every (a, b) in that box, the rational parameters included. This re-run is performed by the macro builder of this document at every build, but it is not yet part of the record, which is why the theorem is stated as it is.

Proposition 1 (calibration)**.** *Let $a = 6$, $b = 0.3$, and let*

$$N_{\pm} = (\pm x_0, 0) + \begin{pmatrix} w & \pm\kappa \\ 0 & H \end{pmatrix}[-1,1]^2, \qquad x_0 = 0.38,\; w = 0.17,\; H = 0.6,\; \kappa = \frac{bH}{12\,x_0} = 0.0395\ldots$$

The instrument certifies the four covering relations $N_{\pm} \overset{F}{\Longrightarrow} N_{\pm}$ and returns the bound $\ln \mathrm{sp}(J) = \ln 2$ with J the all-ones 2×2 matrix (printed 0.693147180560), i.e. $h_{\mathrm{top}}(F_{6,0.3}) \geq \ln 2$, which is the entropy of the full two-shift that Devaney and Nitecki proved to be the nonwandering set of the map for a beyond an explicit threshold depending on b [4].

The proposition is run, as a gate, before the certificate of Theorem 1 is admitted to any page or document of the repository (Section 5).

3 The method

3.1 h-sets and covering relations

An *h-set* is a parallelogram $N = c + A[-1,1]^2$ with $c \in \mathbb{R}^2$ and $A \in \mathbb{R}^{2\times 2}$ invertible; the first column of A is the nominally expanding direction u, the second the nominally contracting direction s. In the coordinates $t = A^{-1}(x - c)$ the set is the square $N_c = [-1,1]^2$, its *exit set* is the pair of u-edges $N_c^- = \{\pm 1\} \times [-1,1]$ and its *entry set* is the pair of lids $N_c^+ = [-1,1] \times \{\pm 1\}$. This is the h-set of Zgliczyński and Gidea [3] with one expanding and one contracting dimension, the coordinate homeomorphism being affine.

Fix two h-sets N_1, N_2 and an integer $k \geq 1$, and let

$$g(t) = A_2^{-1}\big(F^k(c_1 + A_1 t) - c_2\big), \qquad t \in [-1,1]^2,$$

be F^k written from the coordinates of N_1 to those of N_2, $g = (g_u, g_s)$. The instrument certifies the relation $N_1 \overset{F^k}{\Longrightarrow} N_2$ when the following two conditions hold.

(A) *The slab condition.* $g([-1,1]^2)$ does not meet the two closed slabs $\{|u| \le 1,\ |s| \ge 1\}$: every point of the image either has $|g_s| < 1$ or has $|g_u| > 1$.

(B) *The edge condition.* $g_u < -1$ on one of the two u-edges $\{\pm 1\} \times [-1,1]$ and $g_u > 1$ on the other.

Lemma 1. *If (A) and (B) hold then* $N_1 \overset{F^k}{\Longrightarrow} N_2$ *is a covering relation in the sense of [3].*

Proof. The definition in [3] asks for a homotopy $h : [0,1] \times N_{1,c} \to \mathbb{R}^2$ with $h_0 = g$ such that $h_t(N_{1,c}^-) \cap N_{2,c} = \emptyset$ and $h_t(N_{1,c}) \cap N_{2,c}^+ = \emptyset$ for all t, and $h_1(u,s) = (\lambda u, 0)$ with $|\lambda| > 1$. Take, for $t \in [0, \frac{1}{2}]$, $h_t = (g_u, (1-2t)g_s)$: on the exit set $|g_u| > 1$ by (B), so its image never meets $N_{2,c}$; a point of the image with $|g_u| \le 1$ has $|g_s| < 1$ by (A), hence $|(1-2t)g_s| < 1$, so the image never meets a lid. Then, for $t \in [\frac{1}{2}, 1]$, interpolate the u-coordinate linearly from g_u to λu with $\lambda = \pm 2$, the sign chosen so that λu lies on the same side as g_u on each u-edge, and keep $s = 0$: the s-coordinate is never ± 1, and on each u-edge both ends of the interpolation lie strictly beyond ± 1 on the same side, hence so does every point between them. The end map is the linear model. □

The lids alone are not enough: an image that crosses from $u < -1$ to $u > 1$ while passing above the target, with $|g_u| \le 1$ and $g_s > 1$ on the way, satisfies (B) and avoids both lids, yet no point of N_1 lands in N_2 and the homotopy of the lemma does not exist. This is the second defect of Section 6.

3.2 Deciding a relation

Conditions (A) and (B) are decided by evaluating g on sub-boxes of $[-1,1]^2$ in interval arithmetic and bisecting what is not resolved. The arithmetic is the repository's single interval library: every IEEE-754 basic operation returns the correctly rounded nearest double, so widening each computed bound outward by one unit in the last place encloses the exact result; the library is tested against exact rational arithmetic on random operands, and removing the widening turns that test red. The matrix A_2^{-1} is itself enclosed in intervals from the exact determinant formula, so the conditions are verified for the true parallelogram and not for a floating-point approximation of it; a target whose determinant interval contains zero is refused.

For (A) a cell $[u_0, u_1] \times [s_0, s_1]$ is *clear* when the enclosure of g_s over it lies strictly inside $(-1, 1)$ or the enclosure of g_u lies strictly outside $[-1, 1]$; a cell that is not clear is bisected along its longer side. For (B) the same is done on each u-edge, with the enclosure of g_u required to lie strictly beyond -1 or strictly beyond $+1$, the side recorded, and a change of side along one edge refused. The budget is 8,000 cells per condition and a minimum cell width of 1×10^{-6}; exhausting either refuses the relation. For $k > 1$ the map is iterated k times in intervals; the h-sets are small (half-lengths 0.0150 along u and 0.0048 along s, Section 4) and the subdivision is adaptive, so the enclosures stay tight through the composed nonlinearity. Nothing in this procedure can certify a relation for which (A) or (B) fails: every clear cell is clear by a strict inequality on an enclosure of the true image.

3.3 Disjointness

Two parallelograms are proved disjoint by the separating-axis test: along each of the four edge normals, the outward-rounded projections of the two sets are compared, and strict separation along any axis proves disjointness. The test is conservative in the safe direction, since it can fail

to separate sets that are disjoint but never separates sets that intersect. Every pair of h-sets of a certificate is tested, and a pair not proved disjoint refuses the whole certificate, because the itinerary argument below needs the h-sets to separate orbits.

3.4 Chains, itineraries and entropy

The one external theorem consumed is used as a black box, in the way Krawczyk's theorem [5] is consumed by the existence certificates of the same laboratory.

Theorem 2 (Zgliczyński–Gidea [3], as consumed)**.** *Let $(N_t)_{t\in\mathbb{Z}}$ be h-sets and $(f_t)_{t\in\mathbb{Z}}$ continuous maps with $N_t \overset{f_t}{\Longrightarrow} N_{t+1}$ for every t. Then there are points $x_t \in N_t$, $t \in \mathbb{Z}$, with $f_t(x_t) = x_{t+1}$ for every t.*

Let $N_0, \dots, N_{n-1}$ be pairwise disjoint h-sets and $\mathcal{E}$ a set of certified relations (i, j, k), each meaning $N_i \overset{F^k}{\Longrightarrow} N_j$. For an integer $K \geq 1$ define the zero–one matrix

$$B_K[i][j] = 1 \iff \exists\, (a_0, a_1, k_1), \dots, (a_{r-1}, a_r, k_r) \in \mathcal{E} :\ a_0 = i,\ a_r = j,\ k_1 + \cdots + k_r = K.$$

B_K is computed by the recurrence $B_K = \bigvee_k A_k \cdot B_{K-k}$ over the duration matrices A_k of $\mathcal{E}$, with $B_0 = I$, every product read as a boolean product. Nothing is claimed about F^K covering N_j from N_i when $B_K[i][j] = 1$; the entry records the existence of a chain, and the uniform iterate enters only through the times at which itineraries are read.

Lemma 2. *With the notation above, $h_{\mathrm{top}}(F) \geq \frac{1}{K} \ln \mathrm{sp}(B_K)$.*

Proof. Let $\Sigma \subseteq \{0, \dots, n-1\}^{\mathbb{Z}}$ be the set of sequences (i_m) with $B_K[i_m][i_{m+1}] = 1$ for all m, a vertex shift of finite type with $h_{\mathrm{top}}(\sigma|_\Sigma) = \ln \mathrm{sp}(B_K)$ [6, Chapter 4]. Let $\Lambda = \{x : F^{mK}(x) \in \bigcup_i N_i \text{ for all } m \in \mathbb{Z}\}$: it is compact, since F is a homeomorphism and the union is compact, and F^K-invariant. Because the N_i are compact and pairwise disjoint, the index of the h-set containing a point is locally constant on their union, so $\pi : \Lambda \to \{0, \dots, n-1\}^{\mathbb{Z}}$, $\pi(x)_m =$ the index of the h-set containing $F^{mK}(x)$, is well defined and continuous, and $\pi \circ F^K = \sigma \circ \pi$. Given $(i_m) \in \Sigma$, choose for each m a chain of total duration K from i_m to i_{m+1} and concatenate the chains into a bi-infinite sequence of certified relations; Theorem 2 gives points following it, and since each F^k is injective they are the orbit of one point x, which visits N_{i_m} at time mK for every m. Thus $x \in \Lambda$ and $\pi(x) = (i_m)$, so $\Sigma \subseteq \pi(\Lambda)$. The image $\pi(\Lambda)$ is compact and σ-invariant, and a factor has at most the entropy of the system, so $h_{\mathrm{top}}(F^K|_\Lambda) \geq h_{\mathrm{top}}(\sigma|_{\pi(\Lambda)}) \geq h_{\mathrm{top}}(\sigma|_\Sigma) = \ln \mathrm{sp}(B_K)$, the middle inequality by monotonicity in closed invariant subsets. Finally $h_{\mathrm{top}}(F^K|_\Lambda) = K\, h_{\mathrm{top}}(F|_\Lambda)$ [7, Chapter 7], and Λ is F-invariant after replacing it by $\bigcup_{r<K} F^r(\Lambda)$, which does not change the entropy of F^K on it. □

The lemma is where the first defect of Section 6 lived: with mixed durations, two chains that cut the same span K differently (a duration-two relation, or two duration-one relations through an intermediate h-set) can be realised by one and the same orbit, because a duration-two relation constrains nothing at its intermediate time. Counting chains overcounts orbits; B_K is binary precisely so that an itinerary is a sequence of h-sets read at the uniform times mK, where disjointness separates orbits.

3.5 The exact spectral bound

Lemma 3. *Let B be a square matrix with nonnegative integer entries, S a set of indices and B_S the principal submatrix on S. If every row sum of B_S^m is at least $c > 0$, then* $\mathrm{sp}(B) \geq \mathrm{sp}(B_S) \geq c^{1/m}$.

Proof. The spectral radius of a nonnegative matrix dominates that of each principal submatrix [8, Chapter 8]. If $B_S^m \mathbf{1} \geq c\,\mathbf{1}$ entrywise then $B_S^{mr} \mathbf{1} \geq c^r \mathbf{1}$ for every r, so the row-sum norm of B_S^{mr} is at least c^r and $\mathrm{sp}(B_S)^m = \mathrm{sp}(B_S^m) \geq c$ by the spectral radius formula. □

The instrument trims B_K to the set S of indices that keep a positive row sum and a positive column sum within S (iterated to a fixed point), forms the powers B_S^m for $m \leq 8$ in integer arithmetic guarded against exceeding 2^{49} (so that every partial sum stays below 2^{53} and the double arithmetic is exact), and keeps the best value of $\ln(\min_i (B_S^m \mathbf{1})_i)/m$ over m and over $K \leq 40$; the macro builder of this document recomputes the chosen row sum independently in arbitrary-precision integers. The bound is therefore $\frac{1}{K} \ln c^{1/m}$ for an explicit integer c: a logarithm of an integer, never an eigenvalue.

3.6 The proposer

The h-sets and the candidate relations are proposed by a floating-point layer that is believed about nothing. A float orbit of 400,000 points after a transient of 1,000 samples the numerically observed attractor; binning at cell size δ keeps about one centre per occupied cell; each h-set takes its u-direction from the principal axis of the orbit points in its cell and its s-direction from a few backward iterations of the Jacobian, the standard float proposal for the stable direction; a greedy pass keeps a subset proved pairwise disjoint by the certified test. A relation (i, j, k) is nominated when the float image under F^k of the u-segment of N_i passes within $1.6\,\delta$ of the centre of N_j. The hunt tries 6 scales and keeps the best certified bound. The record keeps only what survived certification; the number of nominated relations is not recorded.

3.7 Proof of Theorem 1

The record `certs/entropy-henon.json` holds 340 h-sets with half-lengths 0.0150 along u and 0.0048 along s (the hunt's scale $\delta = 0.03$ times 0.5 and 0.16), and 4,140 relations (i, j, k) with $k \leq 6$. The battery re-derives, from the record alone: that all $\binom{340}{2}$ pairs of h-sets are disjoint by the certified test; that each of the 4,140 relations satisfies (A) and (B) under the procedure of Section 3 within budget; and, by the recurrence, that B_{11} has a core S of 223 indices on which B_S^8 has minimum row sum 338,518,196,978. Lemmas 1, 2 and 3 give $h_{\mathrm{top}}(F) \geq \frac{1}{11} \ln \mathrm{sp}(B_{11}) \geq \frac{1}{88} \ln 338{,}518{,}196{,}978$. □

4 The certificate in numbers

Table 1 lists the relations by duration. Longer durations dominate: the single-step expansion of the map at these parameters averages well below the factor that stretches a cell across a neighbour, so most of the stretching the certificate uses is accumulated over several iterates, with the adaptive subdivision absorbing the composed nonlinearity. The relations connect 4,069 distinct ordered pairs of h-sets; 51 pairs carry more than one duration and 11 relations are loops $N_i \overset{F^k}{\Longrightarrow} N_i$. The mean out-degree is 12.2, the largest out-degree 41 and the largest in-degree

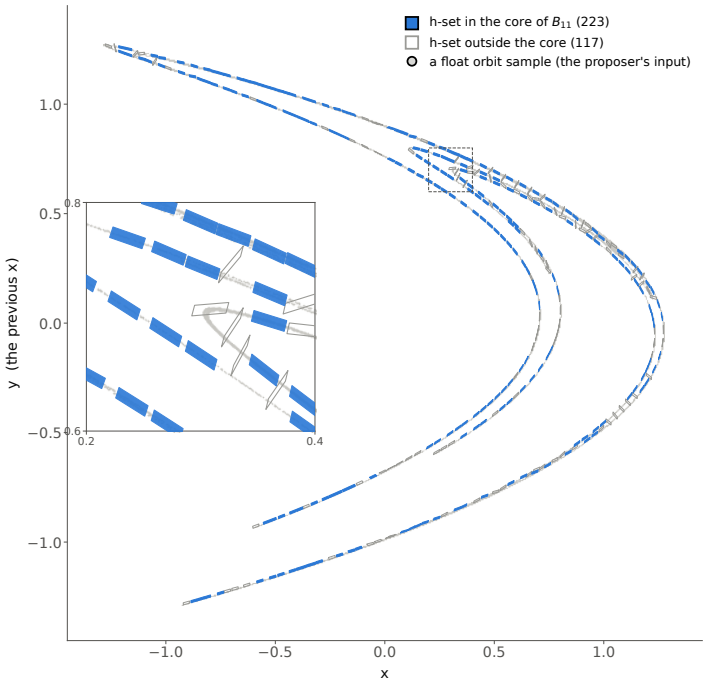


Figure 1: The 340 h-sets of the record, drawn from the certificate: filled, the 223 in the core of B_{11}; outlined, the rest; each a parallelogram with half-lengths 0.0150 and 0.0048. The grey points are a float orbit sample, the proposer's input, for orientation only. The inset is the 0.2-wide window holding the most core h-sets, at scale.

50; 19 h-sets have no outgoing relation and 84 no incoming one, and the union graph over all durations has 115 strongly connected components, one of 226 vertices and the rest singletons. Figure 1 draws the h-sets; Figure 2 draws one relation of each extreme duration in the target's coordinates, and the calibration pair.

duration k	relations $N_i \overset{F^k}{\Longrightarrow} N_j$	cumulative	share
1	101	101	2.4%
2	254	355	6.1%
3	442	797	10.7%
4	751	1,548	18.1%
5	1,099	2,647	26.5%
6	1,493	4,140	36.1%

Table 1: The 4,140 certified covering relations by duration.

Table 2 shows the composed bound for every uniform iterate $K \leq 20$: the number of ordered pairs with a duration-exactly-K chain, the size of the core S, the power m at which the row-sum bound is best and that row sum. The first positive bound appears at $K = 4$; the bound rises to its maximum at $K = 11$ and falls afterwards (0.290851 at $K = 12$, 0.298457 at 13, 0.253596 at 20), because B_K saturates towards the all-ones matrix on its 226-vertex core while the divisor K keeps growing; the instrument's search to $K = 40$ finds nothing above $K = 11$. The entry at

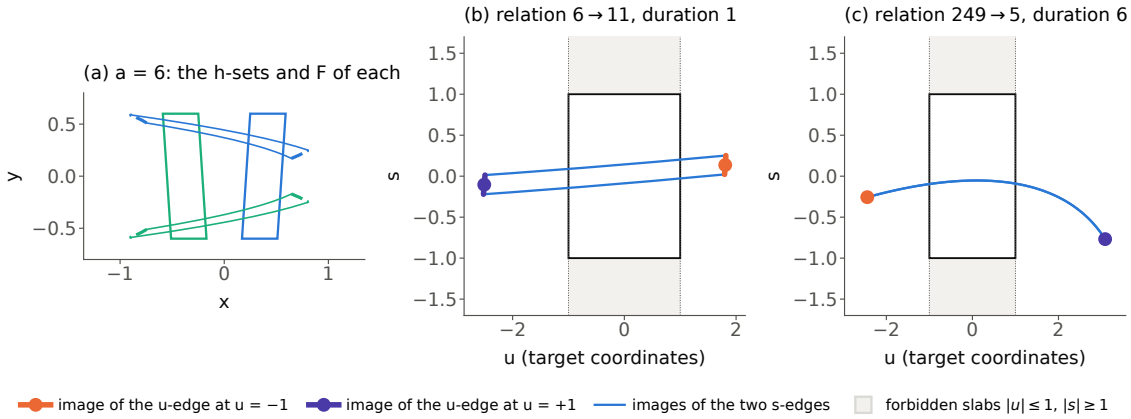


Figure 2: (a) The calibration of Proposition 1 at $a = 6$: the two h-sets and the images of their boundaries under F; each image is a long thin strip that crosses both h-sets, which is the full horseshoe. (b) The first relation of the record, $6 \to 11$ at duration 1, and (c) the duration-6 relation whose image is shortest, $249 \to 5$, both in the target's coordinates: the target is the unit square, the hatched regions are the slabs that condition (A) forbids, the heavy marks are the images of the two u-edges, which condition (B) places strictly beyond $u = \pm 1$ on opposite sides. The curves are float images for the eye; the certificate is the interval evaluation of Section 3.

$K = 11$ is Theorem 1.

The ceiling. The same laboratory's census instrument counts the period-p points of the map exactly: it confines the plane by a certified a priori bound, excludes boxes by interval iteration, resolves the survivors by Krawczyk's operator into boxes holding exactly one periodic point each, and decides identity and minimal period by certificates rather than tolerances; a refusal is possible, a wrong count is not. Its one-off records for $p = 13, \ldots, 16$ at the classical parameters are in Table 3. For a $C^{1+\alpha}$ surface diffeomorphism the exponential growth rate of periodic points dominates the entropy asymptotically [9], which is why the report calls the largest certified rate, 0.4658 at $p = 15$, the *ceiling*: it is where the counts point, and it sits beside the value the report attributes to the literature, 0.4651. Finitely many counts bound nothing, in either direction. The rate at the largest certified period, $p = 16$ with 1,696 points, is 0.4648, the figure the laboratory's handoff notes quote. Theorem 1 is 65% of the ceiling and 44% of $\ln 2$; the gap of 0.1641 nats is the open problem, drawn to scale on the report page.

5 Calibration and controls

The gate of the instrument, `instruments/entropy/battery.js`, runs 12 checks in about 65 seconds and refuses the build on any failure; the macro builder of this document runs it live, so the counts in this section are those of the run that produced the PDF.

1. The calibration of Proposition 1: the two h-sets at $a = 6$, all four relations certified for F itself with a budget of 60,000 cells, the bound equal to $\ln 2$ to twelve digits.

2. Three refusal controls: a target the image never reaches, a target too wide for the stretch, and a target rotated so that the image crosses its lids, must each be refused. All three are

K	pairs with a chain	core $\lvert S\rvert$	m	$\min_i (B_S^m \mathbf{1})_i$	$\frac{1}{mK}\ln(\cdot)$
1	101	1	–	–	0
2	254	4	–	–	0
3	442	21	–	–	0
4	764	118	8	87	0.139560
5	1,218	156	8	624	0.160904
6	2,070	193	8	8,706	0.188995
7	2,651	197	8	477,649	0.233511
8	4,261	204	8	7,989,313	0.248338
9	6,412	216	8	239,701,315	0.267985
10	9,689	217	8	3,681,671,706	0.275333
11	13,740	223	8	338,518,196,978	0.301680
12	19,191	225	8	1,337,304,416,700	0.290851
13	25,542	226	8	30,220,717,975,943	0.298457
14	32,893	226	8	157,232,454,255,130	0.291864
15	40,667	226	7	35,320,068,333,815	0.297100
16	48,158	226	7	76,084,728,649,545	0.285383
17	55,461	226	7	516,229,991,371,881	0.284685
18	61,195	226	6	4,212,458,986,282	0.269158
19	65,937	226	6	5,216,938,231,247	0.256868
20	69,426	226	6	16,451,969,591,183	0.253596

Table 2: The composed bound for every uniform iterate $K \le 20$, recomputed from the recorded relations. A dash means that no power gives a row sum above one there, so the bound is zero.

period p	points of period p, exact	$\ln(N_p)/p$
13	418	0.4643
14	648	0.4624
15	1,082	0.4658
16	1,696	0.4648

Table 3: The certified cycle counts of the census record `certs/census-high-periods.json` (points of period p, i.e. fixed points of F^p, plane exhausted).

refused, and the battery prints the verifier's reason: in the run behind this document all 3 were decided by the edge condition (B) (3 of 3), none by the slab condition (A). This is a finding about the gate: condition (A), the one whose earlier form was the second defect of Section 6, is not exercised by any refusal control today.

3. Two h-sets that overlap must refuse the whole graph; they do.

4. The spectral routine on hand matrices: the full two-graph gives exactly $\ln 2$ at every power, a single two-cycle gives exactly 0, an acyclic graph gives 0.

5. A semantic control for the first defect: the calibration h-sets offered both duration-one and duration-two candidates must still bound by $\ln 2$. The control passes (0.693147181, 4 relations). The macro builder of this document re-runs its eight candidates one by one: the 4 duration-one relations certify and all 4 duration-two candidates are refused, because under F^2 the images of both exit edges of either h-set land on the same side of the target, which is the fold of the horseshoe. The refusals are expected; the consequence is that the

composition of durations is not exercised by this control as it runs, and the first defect is excluded by the construction of B_K rather than by a control that would fire on its return. A control in which duration-two relations do certify is listed in Section 8.

6. The certificate itself: the 340 h-sets re-proved pairwise disjoint, every one of the 4,140 relations re-derived, and the bound recomputed and found no lower than the recorded 0.301680.

6 Two defects found by impossible numbers

Both defects were in the verifier, not in the method, and both were found by the same screen: a lower bound that lands above the value the literature reports for this map is a bug until proved otherwise. Neither was found by reading code. We state them as engineering of the verifier.

Mixed durations counted as itineraries. The first version composed durations by counting: the number of chains of total duration K from i to j was taken as the (i, j) entry of a matrix whose spectral radius was then bounded. As explained after Lemma 2, two chains that cut the same span differently can be realised by one orbit, so the count overstates the number of distinguishable orbits. On the classical map the count returned $h \geq 0.61$ against a literature value near 0.465. The fix is the binary B_K of Lemma 2; the guard is the semantic control of Section 5, with the caveat recorded there.

The lids-only image condition. The first committed version of the verifier required the image to avoid only the two lid segments $\{|u| \leq 1,\ s = \pm 1\}$, the literal entry set, rather than the slabs of condition (A). An image that arches over the target satisfies that condition and (B) without meeting the target, and Lemma 1 has no proof for it. It was found because a two-h-set graph under F at the classical parameters certified with a bound converging to $\ln \varphi = 0.4812$ (φ the golden ratio), above 0.4651. The fix is the slab condition. The record committed under the lids-only condition, `a395c7b4` of 2026-08-25, stated

$$h_{\mathrm{top}}(F) \ \geq \ 0.356403 \qquad \text{(tainted; withdrawn)}$$

from 916 relations of the single uniform iterate F^4 over 228 h-sets; it was replaced the same day by `f86248db`, which carries the certificate of Theorem 1, and the record has 2 versions in its history. The interim number is named here so that it cannot be quoted as a result: its certificate does not establish it, and nothing here decides whether the inequality itself is true.

The lesson the laboratory drew is procedural and is now standing practice: calibrate where the answer is known before any new number counts, and treat a bound above a credible reference value as a detector. The two reference values used here, the literature's 0.4651 and the certified ceiling 0.4658, are not inputs to any proof; they are the alarms.

7 What is not claimed

- *The value.* Nothing here estimates $h_{\mathrm{top}}(F)$; Theorem 1 is a lower bound and the ceiling is a count.
- *An upper bound.* No statement here bounds the entropy from above. The census counts are consistent with the reported literature value but bound nothing at finite period.

- *Priority or improvement.* The author's understanding is that published rigorous lower bounds for this map exceed 0.301680. They are not cited because none is pinned in the record; the claim of this paper is the grammar and the record, not the number.
- *The rational parameters, as recorded.* The record is for the doubles; Remark 2 reports a re-run with one-ulp parameter intervals, performed at every build of this document and not yet part of the record.
- *The attractor.* The h-sets sit near a numerically observed attractor, but the invariant set Λ of Lemma 2 lives inside the h-sets and nothing is claimed about any other invariant set, about hyperbolicity, or about the semi-conjugacy being a conjugacy.
- *Optimality of the certificate.* The proposer used one cell size and one pair of half-lengths for every h-set; the certificate is one point in a large design space.

8 The gap and the next step

The gap of 0.1641 nats to the ceiling has three parts, and the certificate itself says which is cheapest.

The spectral step. The record bounds $\mathrm{sp}(B_{11})$ by a row sum at power $m = 8$. The same matrix supports more: Table 4 continues the exact integer row sums of B_S^m to $m = 32$ by vector iteration (the macro builder of this document computes them; they are not in the record), and a floating-point power iteration on the same core, an indication and not a certificate, puts $\ln \mathrm{sp}(B_S)/11$ near 0.3386. A certified Collatz–Wielandt inequality, $\mathrm{sp}(B_S) \geq \min_i (B_S v)_i / v_i$ for one explicit positive rational vector v evaluated exactly, would recover most of that from the recorded relations alone, with no new interval work. This is the recommended first step.

power m	decimal digits of $\min_i (B_S^m \mathbf{1})_i$	$\frac{1}{m \cdot 11} \ln(\cdot)$
8	12	0.3016
16	25	0.3201
24	38	0.3263
32	51	0.3294

Table 4: The row-sum bound on the recorded core of B_{11} continued past the recorded power, in exact integers (truncated logarithms). The row $m = 8$ is Theorem 1; the rest is not in the record.

The h-sets. All 340 share one shape. Sizing each h-set to the local expansion, admitting longer durations (the share of certified relations grows with k in Table 1), and densifying the core are the recorded next steps of the report; each is new interval work and each is re-proved by the same battery.

The record. Two bookkeeping items follow from this draft: the parameter intervals of Remark 2 should enter the record so that the theorem can be stated for 7/5 and 3/10; and two controls should be added to the battery, one in which duration-two relations do certify so that

the composition of durations is exercised, and one with an image that arches over its target so that condition (A) is the condition that refuses.

The comparison. The published rigorous bounds for this map, and the provenance of the value 0.4651 the report quotes, should be pinned in the corpus and cited; this draft does neither.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The record is `certs/entropy-henon.json` (2 versions; the one in force since `f86248db`, 2026-08-25); the ceiling is `certs/census-high-periods.json`; the verifier is `instruments/entropy/covering.js` and its gate `instruments/entropy/battery.js` (`node instruments/entropy/battery.js`: about 65 seconds, 12 checks, re-deriving every relation of the certificate); the proposer is `tools/hunt-entropy.js` (re-running it rewrites the record); the report page is `reports/entropy.html`, built by `tools/build-report-entropy.js`, which re-verifies the certificate and re-runs the calibration before writing a byte; it is served at `https://carlostoledo.co/reports/entropy.html`, with the record at `https://carlostoledo.co/certs/entropy-henon.json`. The macros of this document are written by `node tools/paper-numbers/henon-entropy.js`, which reads the records, runs the battery, recomputes the composed bound through the instrument and through its own implementation of the recurrence, confirms the row sum in arbitrary-precision integers, performs the one-ulp re-run of Remark 2, and refuses when a record no longer supports a sentence; the figures are drawn by `tools/paper-numbers/henon-entropy-figs.py`; the document is compiled by `node tools/build-paper-tex.js henon-entropy`. Repository: `https://github.com/carlostoledo1891/cert-machine`, concept DOI 10.5281/zenodo.22800699; this build is at commit `aae61990d737`.

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